

Problem 7.42

$$E_d = \sum_{n=-\infty}^{+\infty} |x[n]|^2 = \frac{1}{T} \sum_{n=-\infty}^{+\infty} |x[n]|^2 T = \frac{1}{T} \sum_{n=-\infty}^{+\infty} \int_{nT}^{(n+1)T} |x(nT)|^2 dt$$

$$= \frac{1}{T} \sum_{n=-\infty}^{+\infty} \int_{nT}^{(n+1)T} |x(t)p(t) * h(t)|^2 dt$$

$$= \frac{1}{T} \int_{-\infty}^{+\infty} |x(t)p(t) * h(t)|^2 dt$$

$$= \frac{1}{2\pi T} \int_{-\infty}^{+\infty} \left| \frac{1}{T} \sum_{n=-\infty}^{+\infty} X(j(\omega - \frac{2\pi n}{T})) \cdot T S_a(\frac{\omega T}{2}) e^{-j\frac{\omega T}{2}} \right|^2 d\omega$$

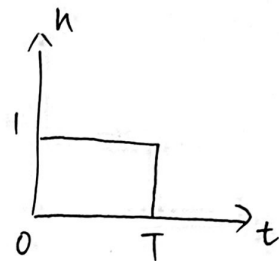
$$= \frac{1}{2\pi T} \int_{-\infty}^{+\infty} \left| \sum_{n=-\infty}^{+\infty} X(j(\omega - \frac{2\pi n}{T})) \right|^2 S_a^2(\frac{\omega T}{2}) d\omega$$

$$= \frac{1}{2\pi T} \int_{-\omega_M}^{\omega_M} |X(j\omega)|^2 \sum_{n=-\infty}^{+\infty} S_a^2(\frac{(\omega + \frac{2\pi n}{T})T}{2}) d\omega$$

$$= \frac{1}{2\pi T} \int_{-\omega_M}^{\omega_M} |X(j\omega)|^2 \sum_{n=-\infty}^{+\infty} \frac{\sin^2(\frac{\omega T}{2})}{(\frac{\omega T}{2} + n\pi)^2} d\omega \leq \frac{1}{2\pi T} \int_{-\omega_M}^{\omega_M} |X(j\omega)|^2 \sum_{n=-\infty}^{+\infty} \frac{1}{(1 + \frac{n\omega_s}{\omega_M})^2} d\omega$$

$$\leq \frac{1}{2\pi T} \int_{-\omega_M}^{\omega_M} |X(j\omega)|^2 \left(\frac{1}{2\pi T} \int_0^{\omega_M} |X(j\omega)|^2 \sum_{n=-\infty}^{+\infty} \frac{1}{(\frac{\omega_s}{\omega} n + 1)^2} d\omega + \int_{-\omega_M}^0 |X(j\omega)|^2 \sum_{n=-\infty}^{+\infty} \frac{1}{(\frac{\omega_s}{-\omega} n - 1)^2} d\omega \right) d\omega$$

$$\leq \frac{1}{2\pi T} \cdot \left(\frac{\pi^2}{8}\right) \cdot E_c = \frac{\pi}{8T} E_c$$



$$H(j\omega) = T S_a(\frac{\omega T}{2}) e^{-j\frac{\omega T}{2}}$$

Problem 7.46

$$x_r[mN] = \sum_{k=-\infty}^{+\infty} x[kN] S_a(\omega_c(M-k)N) = \sum_{k=-\infty}^{+\infty} x[kN] S_a(\pi(M-k)) = x[MN]$$

Problem 7.52

(a) $x(j\omega)p(j\omega) \rightsquigarrow x(t) * p(t) \Leftrightarrow \tilde{x}(j\omega) \Leftrightarrow x(t) * p(t) = \tilde{x}(t)$

(b) $P(j\omega) = \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_0) \Leftrightarrow p(t) = \frac{T}{2\pi} \sum_{k=-\infty}^{+\infty} \delta(t - kT)$, $\tilde{x}(t) = \frac{T}{2\pi} \sum_{k=-\infty}^{+\infty} x(t - kT)$

$$\tilde{x}(t) \text{ and } x(t) = \begin{cases} \frac{T}{2\pi} \cdot \omega \cdot x(t - kT) \Big|_{k=0} = x(t), & |t| < \frac{\pi}{\omega_0} \\ 0, & |t| \geq \frac{\pi}{\omega_0} \end{cases} = x(t)$$

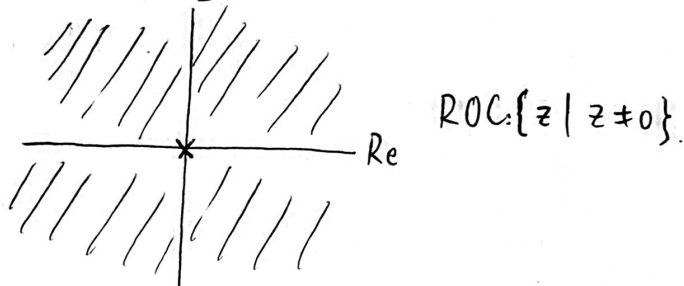
(c) 假设 $t_1 \geq \frac{\pi}{\omega_0}$ 且 $x(t_1) \neq 0$, 记 $t_n = \{\frac{t_1}{T}\}T - \frac{T}{2} = [\{\frac{t_1}{T}\} - \frac{1}{2}]T$, $|t_n| \leq \frac{1}{2}T = \frac{\pi}{\omega_0}$, 不妨假设 $x(t) \geq 0$

$$\tilde{x}(t_n) \geq x(t_n) + x(t_1) \neq x(t_n) \Rightarrow \text{不能还原}$$

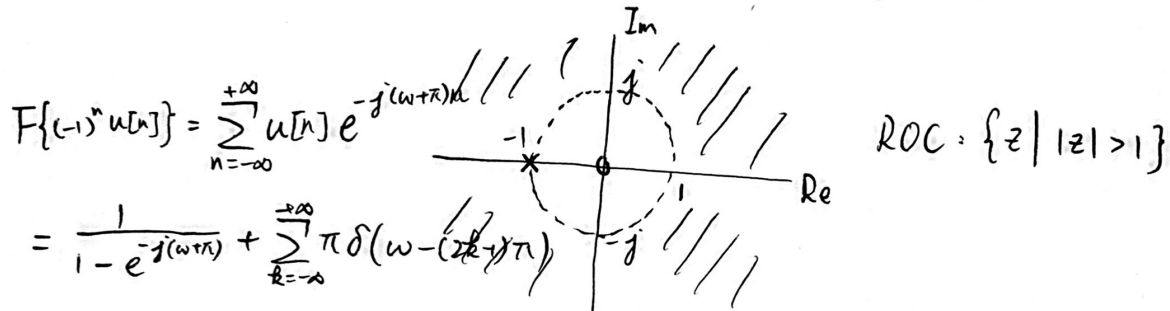
10.21

$$(a) \mathcal{Z}\{\delta[n+5]\} = \sum_{n=-\infty}^{+\infty} \delta[n+5] z^{-n} = z^{-5} \quad \text{Im}$$

$$F\{\delta[n+5]\} = e^{-j(5\omega)}$$

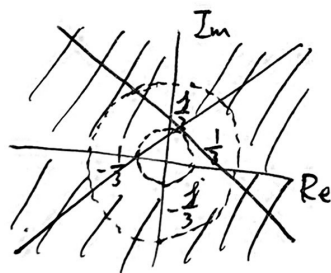
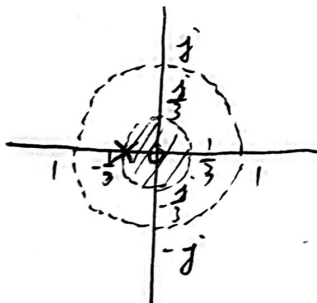


$$(c) \mathcal{Z}\{(-1)^n u[n]\} = \sum_{n=-\infty}^{+\infty} (-1)^n u[n] z^{-n} = \sum_{n=0}^{+\infty} \left(-\frac{1}{z}\right)^n = \frac{1}{1 + \frac{1}{z}} = \frac{z}{z+1}$$



$$(d) \mathcal{Z} = \frac{1}{1 + e^{-j\omega}} + \sum_{k=-\infty}^{+\infty} \pi \delta(\omega - (2k-1)\pi)$$

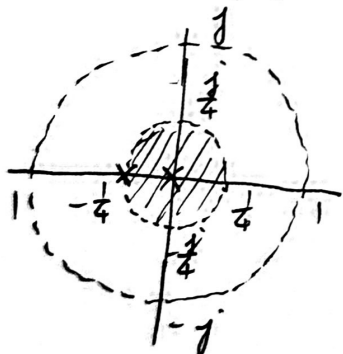
$$(e) \mathcal{Z}\left\{\left(-\frac{1}{3}\right)^n u[-n-2]\right\} = \sum_{n=-\infty}^{-2} \left(-\frac{1}{3}\right)^n z^{-n} = \sum_{n=-\infty}^{-2} (-3z)^{-n} = \sum_{n=2}^{+\infty} (-3z)^{-n} = 9z^2 \cdot \frac{1}{1+3z} = \frac{9z^2}{1+3z}$$



$$\text{ROC: } \{z \mid |z| < \frac{1}{3}\}$$

$$F\left\{\left(\frac{1}{3}\right)^n u[-n-2]\right\} \text{ 不存在}$$

$$(f) \mathcal{Z}\left\{\left(\frac{1}{4}\right)^n u[3-n]\right\} = \sum_{n=-\infty}^3 \left(\frac{1}{4}\right)^n z^{-n} = \sum_{n=-3}^{+\infty} (4z)^n = (4z)^{-3} \frac{1}{1-4z} = \frac{1}{(4z)^3(1-4z)}$$



$$\text{ROC: } \{z \mid 0 < |z| < \frac{1}{4}\}, \quad F\left\{\left(\frac{1}{4}\right)^n u[3-n]\right\} \text{ 不存在}$$